Susceptibility of Fermi gas

$$\chi_0(\vec{r},t) = -i\theta(t)\left\langle [\hat{\rho}(\vec{r},t), \hat{\rho}(0,0)]\right\rangle_{H_0}$$

$$\hat{\rho}(\vec{r}) = \hat{\psi}^+(r)\hat{\psi}(r) = \frac{1}{V}\sum_{kk'} \hat{a}^+_k \hat{a}_{k'} e^{-i(\vec{k}-\vec{k}')\vec{r}}$$

$$\hat{H}_0 = \sum_k \left(\frac{\hbar^2 k^2}{2m} - \mu\right) a^+_k a_k$$

$$\hat{\rho}(\vec{r},t) = e^{i\hat{H}_0 t/\hbar}\,\hat{\rho}(\vec{r})\,e^{-i\hat{H}_0 t/\hbar} =$$

$$= \frac{1}{V}\sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}} \underbrace{e^{i\hat{H}_0 t/\hbar}\,\hat{a}^+_k\,e^{-i\hat{H}_0 t/\hbar}}_{\hat{a}^+_k(t)} \underbrace{e^{i\hat{H}_0 t/\hbar}\,\hat{a}_{k'}\,e^{-i\hat{H}_0 t/\hbar}}_{\hat{a}_{k'}(t)}$$

$$i\hbar\,\partial_t \hat{A}(t) = [\hat{A}, \hat{H}]$$

$$\Rightarrow i\hbar\,\partial_t \hat{a}_{k'}(t) = [\hat{a}_{k'}, \hat{H}_0] = \sum_k \left(\frac{\hbar^2 k^2}{2m} - \mu\right)[\hat{a}_{k'}, \hat{a}^+_k a_k]$$

Fermions

$$\underbrace{\hat{a}_{k'}\hat{a}^+_k}_{\delta_{kk'} - \hat{a}^+_k \hat{a}_{k'}}\hat{a}_k - \hat{a}^+_k \hat{a}_k \hat{a}_{k'} = \delta_{kk'}\hat{a}_k - \hat{a}^+_k \underbrace{\hat{a}_{k'}\hat{a}_k}_{-\hat{a}_k \hat{a}_{k'}} - \hat{a}^+_k \hat{a}_k \hat{a}_{k'}$$

$a_{k'}a^+_k + a^+_k a_{k'} = \delta_{kk'}$

$$= \delta_{kk'}\hat{a}_k$$

$$\Rightarrow i\hbar\,\partial_t \hat{a}_{k'}(t) = \sum_k \delta_{kk'}\left(\frac{\hbar^2 k^2}{2m} - \mu\right)\hat{a}_k(t) =$$

$$= \left(\frac{\hbar^2 k'^2}{2m} - \mu\right)\hat{a}_{k'}(t) = (\varepsilon_{k'} - \mu)\,\hat{a}_{k'}(t)$$

$$\Rightarrow \hat{a}_{k'}(t) = e^{-i(\varepsilon_{k'}-\mu)t/\hbar}\,\hat{a}_{k'}(0)$$

$$\hat{a}_k^+(t) = \left(\hat{a}_k(t)\right)^+ = \left(e^{iH_0 t/\hbar}\, \hat{a}_k\, e^{-iH_0 t/\hbar}\right)^+ =$$

$$= e^{iH_0 t/\hbar}\, \hat{a}_k^+\, e^{-iH_0 t/\hbar}$$

$$\Rightarrow \hat{a}_k^+(t) = \left(e^{-i(\varepsilon_k-\mu)t/\hbar}\, \hat{a}_k(0)\right)^+ = e^{i(\varepsilon_k-\mu)t/\hbar}\, \hat{a}_k^+$$

Note $\hat{a}_{k'}(0) = \hat{a}_{k'}$; $\hat{a}_k^+(0) = \hat{a}_k^+$.

$$\Rightarrow \hat{\rho}(\vec{r},t) = \frac{1}{V} \sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}}\, e^{i(\varepsilon_k-\mu)t/\hbar}\, e^{-i(\varepsilon_{k'}-\mu)t/\hbar} \cdot \hat{a}_k^+ \hat{a}_{k'}.$$

$$\chi_0(\vec{r},t) = -i\theta(t)\, \frac{1}{V} \sum_{kk'} e^{-i(\vec{k}-\vec{k}')\vec{r}}\, e^{i(\varepsilon_k-\mu)t/\hbar}\, e^{-i(\varepsilon_{k'}-\mu)t/\hbar}$$

$$\cdot \left\langle \left[\hat{a}_k^+ \hat{a}_{k'}, \underbrace{\hat{\rho}(0,0)}_{\frac{1}{V}\sum_{k_1,k_2} \hat{a}_{k_1}^+ \hat{a}_{k_2}}\right]\right\rangle$$

Need to evaluate

$$[a_k^+ a_{k'}, a_{k_1}^+ a_{k_2}] = a_k^+ a_{k'} a_{k_1}^+ a_{k_2} - a_{k_1}^+ a_{k_2} a_k^+ a_{k'}$$

$$a_k^+ \left(\delta_{k'k_1} - a_{k_1}^+ a_{k'}\right) a_{k_2} = \delta_{k'k_1}\, a_k^+ a_{k_2} - a_k^+ a_{k_1}^+ a_{k'} a_{k_2}$$

$$= \delta_{k'k_1}\, a_k^+ a_{k_2} - a_{k_1}^+ a_k^+ a_{k_2} a_{k'} =$$

$$= \delta_{k'k_1}\, a^+_k a_{k_2} - a^+_{k_1}\left(\delta_{kk_2} - a_{k_2} a^+_k\right) a_{k'} =$$

$$= \delta_{k'k_1}\, a^+_k a_{k_2} - \delta_{kk_2}\, a^+_{k_1} a_{k'} + a^+_{k_1} a_{k_2} a^+_k a_{k'}$$

Thus $\langle [\xi, \,] \rangle = \underbrace{\langle a^+_k a_{k_2} \rangle}_{\langle a^+_k a_k \rangle \delta_{kk_2}} \delta_{k'k_1} - \underbrace{\langle a^+_{k_1} a_{k'} \rangle}_{\langle a^+_{k'} a_{k'} \rangle \delta_{k_1 k'}} \delta_{kk_2}$

$$\chi_0(\vec{r}, t) = -i\,\theta(t)\, \frac{1}{V^2} \sum_{\substack{kk' \\ k_1 k_2}} \left(n_k\, \delta_{kk_2} \delta_{k'k_1} - n_{k'}\, \delta_{k'k_1} \delta_{kk_2} \right)$$

$$\cdot\, e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}}\, e^{i(\varepsilon_k - \mu)t/\hbar}\, e^{-i(\varepsilon_{k'} - \mu)t/\hbar}$$

Now recall that

$$\chi(\vec{q}, \omega) = \int d\vec{r}\, dt\, e^{-i\vec{q}\cdot\vec{r} + i\omega t}\, \chi(\vec{r}, t)$$

So that two integrals are needed

$$\int d\vec{r}\, e^{-i\vec{q}\cdot\vec{r}}\, e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}} = \int d\vec{r}\, e^{-i\vec{r}\cdot(\vec{q}+\vec{k}-\vec{k}')}$$

$$= V \cdot \delta_{\vec{q}+\vec{k}-\vec{k}',\,0} \quad \Rightarrow \vec{k}' = \vec{k} + \vec{q}$$ [here we think of finite volume system

$$\int_{-\infty}^{\infty} dt\, (-i\,\theta(t))\, e^{it\left(\frac{\varepsilon_k - \mu}{\hbar} - \frac{\varepsilon_{k'} - \mu}{\hbar}\right)}\, e^{i\omega t} =$$

$$= -i \int_0^{\infty} dt\, e^{it\left(\omega + i\delta + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar}\right)} = \Big(\delta = +0 \text{ comes from } \theta(t)\Big)$$

$$= (-i) \quad \frac{e^{it\left(\omega + i\delta + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar}\right)}\Big|_0^\infty}{i\left(\omega + i\delta + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar}\right)} = (-i)\,\frac{-1}{i\left(\omega + \frac{\varepsilon_k - \varepsilon_{k'}}{\hbar} + i\delta\right)}$$

Collect together

$$\chi(\vec{q},\omega) = \frac{1}{V}\sum_{\vec{k}} \frac{n_{\vec{k}} - n_{\vec{k}+\vec{q}}}{\omega + i\delta + \frac{1}{\hbar}\left(\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}+\vec{q}}\right)}$$

$$n_{\vec{k}} \equiv \langle \hat{a}^+_k \hat{a}_k \rangle$$

$$\varepsilon_{\vec{k}} = \begin{cases} \dfrac{\hbar^2 \vec{k}^2}{2m} & \text{for free system of fermions;} \\ \text{can be anything} \rightarrow \text{we have not used its explicit form yet!} \end{cases}$$

Moreover, $\varepsilon_k - \varepsilon_{k+q} = (\varepsilon_k - \mu) - (\varepsilon_{k+q} - \mu)$.

(And we set $\hbar = 1$ now — it can be restored at any stage if necessary)

Static susceptibility, $\omega = 0$:

$$\chi(\vec{q},0) = \frac{1}{V}\sum_k \frac{n_k - n_{k+q}}{i\delta + \varepsilon_k - \varepsilon_{k+q}}$$

Using $\mathrm{Im}\left(\frac{1}{x+i\delta}\right) = \mathrm{Im}\left(\frac{P}{x} - i\pi\delta(x)\right) = -\pi\delta(x)$

$$\mathrm{Im}\,\chi(\vec{q},0) = -\pi \frac{1}{V}\sum_k (n_k - n_{k+q})\,\delta(\varepsilon_k - \varepsilon_{k+q})$$

But $n_k = \langle a_k^+ a_k \rangle = f(\varepsilon_k) = \frac{1}{e^{\frac{\varepsilon_k - \mu}{k_B T}} - 1}$ (Fermi-Dirac dist.)

$$\mathrm{Im}\,\chi(\vec{q},0) = -\pi \frac{1}{V}\sum_k \left(f(\varepsilon_k) - f(\varepsilon_{k+q})\right)\delta(\varepsilon_k - \varepsilon_{k+q})$$

$\delta(\varepsilon_k - \varepsilon_{k+q})$ requires $\varepsilon_k = \varepsilon_{k+q}$ which makes the two Fermi functions identical, $f(\varepsilon_k) = f(\varepsilon_{k+q}) \Rightarrow \underline{\mathrm{Im}\,\chi(\vec{q},0) = 0}$

Static susceptibility is real — quite obvious statement as there can not be energy absorption from the static field.

$$\chi(\vec{q},0) = \mathrm{Re}\left(\chi(\vec{q},0)\right) = \frac{1}{V}\sum_k \frac{n_k - n_{k+q}}{\varepsilon_k - \varepsilon_{k+q}}$$

Consider now uniform $\vec{q} \to 0$ limit: convenient to write everything in terms of $\xi_k = \varepsilon_k - \mu$

$$\chi(\vec{q}\to 0, \omega = 0) = \frac{1}{V}\sum_k \frac{f(\xi_k) - f(\xi_{k+q})}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{f(\xi_k) - f(\xi_k + \xi_{k+q} - \xi_k)}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{f(\xi_k) - \left[f(\xi_k) + (\xi_{k+q} - \xi_k) \frac{\partial f}{\partial \xi_k} \right]}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{V} \sum_k \frac{\partial f(\xi_k)}{\partial \xi_k} = \int d\xi \, \underbrace{N(\xi)}_{\text{density of states}} \frac{\partial f}{\partial \xi} = -N(0)$$

Density of states at the Fermi level.

Since $\frac{\partial f}{\partial \xi} \xrightarrow[T \to 0]{} -\delta(\xi)$ (this is important property of Fermi distribution function)

Now treat general situation with $\omega \neq 0$:

$$\mathrm{Im}\left(\chi(\vec{q},\omega)\right) = -\frac{\pi}{V}\sum_{k}\left(f(\xi_k)-f(\xi_{k+q})\right)\delta\left(\omega+\xi_k-\xi_{k+q}\right)$$

$$\xi_k = \frac{k^2}{2m}-\mu = \frac{k^2-k_F^2}{2m}$$ where for spherical Fermi surface $\mu = \frac{k_F^2}{2m}$

$$\xi_k-\xi_{k+q} = \frac{\vec{k}^2-k_F^2}{2m}-\frac{(\vec{k}+\vec{q})^2-k_F^2}{2m} = -\left(\frac{\vec{k}\cdot\vec{q}}{m}+\frac{\vec{q}^2}{2m}\right)$$

Consider $\omega>0 \Rightarrow \delta(\ )$ dictates $\omega = \frac{\vec{k}\cdot\vec{q}}{m}+\frac{q^2}{2m}>0$

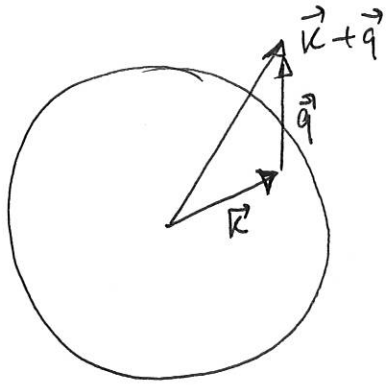
But $\vec{k}\cdot\vec{q} = kq\cos\theta_{kq} \in (-kq, +kq)$

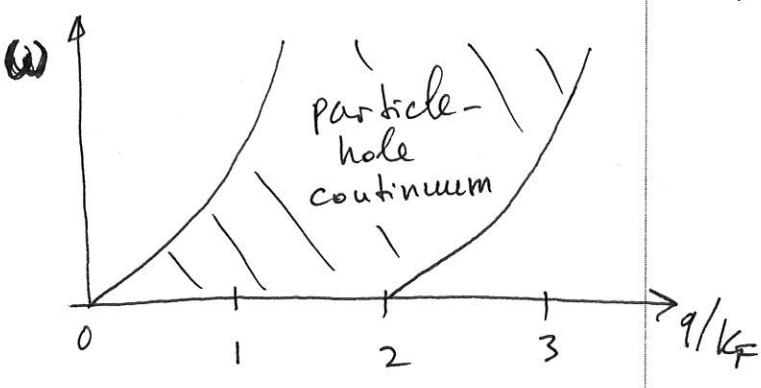
$$\Rightarrow \begin{cases} \omega_{max} = \frac{kq}{m}+\frac{q^2}{2m} \\ \omega_{min} = -\frac{kq}{m}+\frac{q^2}{2m} \end{cases}$$

<u>T=0</u> Sharp Fermi functions $\Rightarrow f(\xi_k)-f(\xi_{k+q}) =$

$$= f(\xi_k)-f(\omega+\xi_k) \neq 0 \text{ if } \begin{cases} k>k_F,\ |\vec{k}+\vec{q}|<k_F & \omega<0 \\ k<k_F,\ |\vec{k}+\vec{q}|>k_F & \omega>0 \end{cases}$$

<u>$\omega>0$</u>





ω_{max} for $\vec{q}\to 0$: $\omega_{max} = v_F q + q^2/2m$ $\left(v_F = \frac{k_F}{m}\right)$

ω_{min} requires $q>2k_F$: $\omega_{min} = -v_F q + \frac{q^2}{2m}$

Energy is absorbed inside shaded area: corresponds to particle $\to$ hole excitation

$$\chi(\vec{q},\omega) = \frac{1}{V}\sum_k \frac{f(\xi_k) - f(\xi_{k+q})}{\omega + \xi_k - \xi_{k+q} + i\delta} =$$

For small, but finite, $\vec{q}$

$$= \int \frac{d\vec{k}}{(2\pi)^3} \frac{f(\xi_k) - f(\xi_{k+q})}{\omega - \frac{\vec{k}\cdot\vec{q}}{m} - \frac{q^2}{2m} + i\delta} \cong \underbrace{\int d\varphi}_{2\pi} \int_0^\pi d\theta \sin\theta \int \frac{dk\, k^2}{(2\pi)^3}$$

$$\cdot \frac{f(\xi_k) - f(\xi_{k+q})}{\omega - V_F q\cos\theta + i\delta} \quad , \; V_F = \frac{k_F}{m}$$

$$f(\xi_{k+q}) = f(\xi_k + \xi_{k+q} - \xi_k) \approx f(\xi_k + V_F q\cos\theta)$$

$$\delta\xi = \xi_{k+q} - \xi_k = \frac{(\bar{k}+\bar{q})^2}{2m} - \frac{k^2}{2m} \approx \frac{\bar{k}\cdot\bar{q}}{m} = V_F q\cos\theta$$

At $T=0$ $\quad f(\xi_k) - f(\xi_k + \delta\xi) = \begin{cases} (\xi_k < 0,\ \xi_k + \delta\xi > 0) & 1 \\ (\xi_k > 0,\ \xi_k + \delta\xi < 0) & -1 \end{cases}$

$$= \begin{cases} 1, & \cos\theta > 0 \text{ and } \xi_k < 0 \\ -1, & \cos\theta < 0 \text{ and } \xi_k > 0 \end{cases}$$

Write $f(\xi_k) - f(\xi_{k+q}) \approx f(\xi_k) - \left[f(\xi_k) + \delta\xi \frac{\partial f(\xi_k)}{\partial \xi_k}\right] =$

$$= -\delta\xi \frac{\partial f(\xi_k)}{\partial \xi_k} = -V_F q\cos\theta \frac{\partial f}{\partial \xi}$$

$$\chi(\vec{q},\omega) \cong \int \frac{dk\, k^2}{(2\pi)^2} \int_0^\pi d\theta \sin\theta \frac{-V_F q\cos\theta \frac{\partial f(\xi_k)}{\partial \xi_k}}{\omega - V_F q\cos\theta + i\delta}$$

$\frac{\partial f}{\partial \xi} \sim -\delta(\xi)$ is sharply peaked at $\xi_k = 0$ where

$$\xi_k = V_F K$$

$$\Rightarrow \int dk\, k^2 \ldots = \int d\xi \frac{K_F^2}{V_F} \ldots = (m K_F) \int d\xi \ldots$$

Also, $\int_{-\infty}^{\infty} d\xi \frac{\partial f}{\partial \xi} = f(\xi)\Big|_{-\infty}^{\infty} = \frac{1}{e^{\beta\xi}+1}\Big|_{-\infty}^{\infty}$

$= 1.$

$$\Rightarrow \chi(\vec{q}, \omega) = \frac{m K_F}{(2\pi)^2} \int_0^{\pi} d\theta \sin\theta \frac{V_F q \cos\theta}{\omega - V_F q \cos\theta + i\delta}$$

$$= \frac{1}{2} N_0 \int_{-1}^{1} d(\cos\theta) \frac{\cos\theta}{\frac{\omega}{V_F q} - \cos\theta + i\delta}$$

where $N_0 = \frac{m K_F}{2\pi^2}$ is density of states at the Fermi level.

Denote $\cos\theta = z$, $\frac{\omega}{V_F q} = z_0$

$$\int_{-1}^{1} dz \frac{z}{z_0 - z + i\delta} = -\int_{-1}^{1} dz \frac{z_0 - z - z_0}{z_0 - z + i\delta} =$$

$$= -\int_{-1}^{1} dz \left(1 - \frac{z_0}{z_0 - z + i\delta}\right) = -2 - z_0 \ln(z_0 - z + i\delta)\Big|_{-1}^{1} =$$

$$\cancel{= -2 - z_0 \ln \frac{z_0 - 1 + i\delta}{z_0 + 1 + i\delta}} = -2 - z_0 \ln \frac{z_0 - 1}{z_0 + 1}$$

$$= -2 + z_0 \ln \frac{z_0 + 1 + i\delta}{z_0 - 1 + i\delta}.$$

So, $\chi(\vec{q},\omega) = \frac{1}{2} N_0 \left\{ -2 + \frac{\omega}{v_F q} \ln\left(\frac{\omega + v_F q + i\delta}{\omega_1 - v_F q + i\delta} \right) \right\}$

Getting Imaginary part of the log can be a bit tricky but we can make few simple observations:

for $\omega \gg v_F q$ the argument of log is positive
$\Rightarrow$ no Im part.

for $\omega < 0$ but large so that again $|\omega| \gg v_F q$
Im part is absent again.

Then go back to integral form:

$$\mathrm{Im}\,\chi = \frac{1}{2} N_0 \int_{-1}^{1} dx \; x \, (-\pi) \, \delta(z_0 - x) \quad (x = \cos\theta)$$

$$= -\frac{\pi}{2} N_0 z_0 \quad \text{provided that } -1 < z_0 < 1$$

$$\Rightarrow \mathrm{Im}\,\chi(\vec{q},\omega) = -\frac{\pi}{2} N_0 \frac{\omega}{v_F q} \, \theta\left(1 - \frac{|\omega|}{|v_F q|}\right)$$

$$\theta(y) = \begin{cases} 1, & y > 0 \\ 0, & y \le 0 \end{cases} = \text{step function.}$$

Thus $\chi(\vec{q},\omega) = N_0 \left\{ -1 + \frac{\omega}{2 v_F q} \ln \left| \frac{\omega + v_F q}{\omega - v_F q} \right| \right.$

$$\left. - i\pi \frac{\omega}{2 v_F q} \, \theta\left(1 - \left|\frac{\omega}{v_F q}\right|\right) \right\}$$